

IT640 - Telecommunications/Networking 2025

9-1 Final Project Submission: Network Analysis and Architecture
Evaluation

SNHUEnergy, Inc.

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Introduction - This report is about the network architecture of SNHUEnergy Inc. which is an oil and gas organization. The company operates from two main locations, Dallas (HQ, 90 employees) and Memphis (30 employees) and are connected using a wide area network (WAN). Primary internet gateway, application, and firewall (security controls) are provided by the Dallas office and Memphis depends on this WAN link for any external connectivity. SNHUEnergy heavily relies on the communication between the two locations for day to day operations. The company plans to expand its operations into Houston and Kansas City growing its employees by 50% annually for next two years. To support long term growth, the company requires a secure and scalable network architecture.

Executive Summary

The network evaluation and analysis identified that the data traffic between Dallas and Memphis offices experiences bottlenecks and latency, especially when accessing shared resources like server farms, video conferencing and enterprise applications. The current security posture does not offer ample protection across internal nodes, which increases risk exposure in the network. To improve security and performance this report outlines key modifications like VPN based remote connection, redundant firewalls, and centralized network monitoring for threat detection and visibility in real time. Migrating core services/applications such as HR, payroll, and communication systems to cloud platforms helps improve uptime and reduces hardware dependency. Additionally, the implementation of VoIP solution over SD-WAN ensures secure, high quality communication and business continuity throughout the SNHUEnergy offices.

Current Network Architecture

Network Applications - A network is a connection between two or more devices that communicate to share data and resources. As Cummings (2024) explains, it “links other

subsystems together and allows them to share information and resources” (para. 1). The Dallas office network applications include email, payroll, accounting, and HR systems. Meanwhile the Memphis office network applications include billing and operations applications. At the organizational level, the two common services used by both offices are VoIP phone system and video conferencing tools. The applications mentioned rely on domain name system (DNS) for name resolution, dynamic host configuration protocol (DHCP) for assigning IP addresses, and network time protocol (NTP) for time synchronization. These essential network services ensure seamless connectivity and consistent performance between offices.

Table - OSI Model Identification and Label

OSI Model Layers with brief description	SNHUEnergy Network Components
Layer 7 - Application (Closest to user, provides application services)	Email, Payroll, Accounting, HR, Billing, Operations, VoIP, Video conferencing
Layer 6 - Presentation (Encrypts, formats and compress usable data)	TLS/SSL for secure email, VPN for remote access, file compression and formatting (e.g., PDF, DOCX, CSV used in HR/Payroll systems)
Layer 5 - Session (Manages communication sessions)	Servers, web browsers, and applications mentioned in application layer
Layer 4 - Transfer (Ensures reliable data transfer using TCP & UDP)	Firewall (Dallas only), TCP/UDP protocols supporting VoIP, video conferencing, and apps
Layer 3 - Network (Assigns global addresses to interfaces and finds best	Firewall(Dallas only), Routers - to manage routing and IP addressing between subnets, in future the

routes)	configurations will include layer 3 paths as redundancy (primary and secondary LAN)
Layer 2 - Datalink (Framing, physical addressing MAC, error, flow and access control)	Switches, Network interface cards, and Wireless access points(Dallas only)
Layer 1 - Physical (Transmits raw bits, responsible for actual physical connection between devices)	WAPs(Dallas only), Ethernet/fiber cables, Connectors(RJ-45)

Note: Computers are components of all OSI layers because they have hardware, operating system, network card, and applications. Firewall filters the traffic for both offices, since the Memphis office is not directly linked with the internet and connects with the internet by the Dallas office. The company's growth needs future planning, and for that we will define structured IP schemes and subnets to improve management and redundancy across switches and routers.

Key Component Descriptions - Some of the key components in the SNHUEnergy network are routers, switches, firewall, wireless access points, computers and servers. A router connects different networks and directs data using IP addresses. In this network it connects the Dallas office to Memphis and it also connects to the Internet. A switch is a vital component in LANs, used to transmit data between connected devices. As Juniper Networks explains, “A network switch connects users, applications, and equipment across a network so that they can communicate with one another and share resources.” In this network the Dallas office has two switches and the Memphis office has one, all three connected to their respective routers. A

Firewall protects the network and monitors traffic. According to Cisco (n.d.), “A firewall is the first line of defense that monitors incoming and outgoing traffic and decides to allow or block specific traffic based on a defined set of security rules.” The Dallas office includes one firewall which is placed between the internet and the router to filter out traffic. Wireless Access Points is another key component in the network, whose main function is to provide wireless connectivity for mobile devices within the office. In simple terms, it provides the ability to share resources between devices without using cables. The Dallas office has three WAPs connected to switch1 in the network. Computers allow users to interact and access the applications on the servers.

Network Analysis - SNHUEnergy has two offices, Dallas and Memphis, which are linked together via their routers as a Wide Area Network (WAN). The Memphis office relies on the connection to the Dallas office as it is not connected directly to the internet, so everything goes through the Dallas office. The Dallas office can survive on its own but in case of router failure at the Memphis office, it will not be able to access the applications residing in the Memphis office, such as billing and operations. SNHUEnergy as an organization heavily relies on the network connection between the two offices as they share various services/applications across them which are essential for the organization to function. There is only one router on each location which means there is no redundancy and a single point of failure will disrupt the whole network, and it would put the Memphis office at standstill. Memphis office has only one router and one switch, so again single point of failure at layer 2 and layer 3. The Dallas office however has two switches so it provides redundancy at layer 2, but only one router so no redundancy at layer 3. There is only one firewall which sits between the internet and the router at the Dallas office, monitoring inbound/outbound traffic, and filtering by the set security rules. Additionally, there is no protection from internal threats in either office. To mitigate these risks, the company should

implement a redundant backbone by creating primary and secondary LAN paths between both sites. Firewall should be deployed at each site for security. In the firewall the use of intrusion detection system (IDS) and intrusion prevention system (IPS) is crucial, because they acknowledge and allow/block the packets based on the policy the administrator sets in the firewall. Since the Memphis office is not connected to the internet directly, all the traffic comes through the Dallas office which is filtered by the firewall there. WAP's in the Dallas office connects all the devices to each other wirelessly. Additional links should be added to increase redundancy between routers and switches to prevent single point of failure in the network. In the coming sections we will discuss the IP addressing schemes including subnet ranges, dynamic routing protocol (such as open shortest path first (OSPF)) redundant LANs, IDS and IPS to improve security, scalability and resiliency.

Physical Network Devices - In this section I will discuss the physical devices in the network that are used to connect various types of devices within the organization, and I will also discuss some of the devices that we will need to expand the organization to two new locations. The Dallas office has a router, two switches, a firewall, three wireless access points (WAPs) and the Dallas office is directly connected to the internet. Meanwhile the Memphis office has a router, and a switch making it less redundant and accesses the internet via the Dallas office which means it totally relies on the Dallas office to maintain the connectivity. For the new offices we will need very similar infrastructure with a router, a firewall, a couple switches and WAPs according to the staff size. Physical network devices also include management hosts for remote access and servers, such as Application and MySQL Database Servers in Dallas. The company should also integrate Remote Authentication Dial-In User Service (RADIUS) authentication server and Active Directory(AD)/Lightweight Directory Access Protocol (LDAP) for centralized user

management. The LDAP is used for Single Sign On (SSO) inside the firewall, which authenticates users and registers their account in the firewall. If an unauthenticated user tries to connect, the firewall inspects it and decides whether or not to allow it. The Memphis office has VoIP endpoints for voice and video conferencing.

Critical Traffic Patterns - The Traffic Flow Wireshark Capture document shows various types of traffic and protocols being used throughout which include Real-Time Transfer Protocol (RTP), Transmission Control Protocol (TCP), Secure Shell (SSH) and MySQL. In this section I will breakdown the traffic flow captured in the document using the mentioned protocols.

Service (VoIP) - RTP is used for video conferencing and VoIP. According to Spiceworks, “Real-Time Transport Protocol (RTP) is defined as a network communication protocol developed on the User Datagram Protocol (UDP), which helps transport audio, video, and media traffic in real-time while minimizing jitter and packet loss” (Spiceworks, n.d.). The Wireshark packets that are captured show many RTP packets such 2198, 2202, 2204 etc. which show continuous flow of RTP traffic from source IP 67.16.104.172 to destination IP 10.0.6.73.

Application (TCP/MySQL) - TCP is a connection-oriented protocol which uses three-way handshake (SYN, SYN-ACK, ACK) to establish a session before transferring data. TCP uses flow control and congestion control to manage traffic so the receiver is not overwhelmed and sending rate is adjusted based on the traffic. TCP has overhead and latency but is very reliable unlike UDP. According to Fortinet, “Transmission Control Protocol (TCP) is a communications standard that enables application programs and computing devices to exchange messages over a network” (Fortinet, n.d.). The Wireshark packets 2197, 2203, 2210, 2217, and 2221 show MySQL transactions between 10.0.8.73 (source IP - application server) and 10.0.8.42

(destination IP - database server) on TCP port 3306. The other TCP transactions are mostly for SSH and routine TCP control.

MySQL is a relational database management system (RDBMS) which companies use to store data, which can then be extracted using Structured Query Language (SQL), which is the most common language to extract and organize data from databases such as MySQL. The Wireshark capture shows that the application server uses ephemeral ports (e.g. 52260/52261) and connects to the database server on port 3306. Both source and destination are on 10.0.8.0/24 and short ACKs show very low latency. Another thing that I noticed is that there is no transaction using TLS which means that the transactions are likely unencrypted. The Wireshark also shows successful responses which indicates that queries are completing normally.

Network Management (SSH) - Secure Shell (SSH) is a secure and encrypted remote access protocol which is used to manage systems over networks (TCP/22). As Kerner notes, “The Secure Shell (SSH) protocol serves as the backbone of modern network administration, providing encrypted remote access to virtually every server, network device and embedded system in enterprise environments” (Kerner, 2025). The Wireshark capture shows SSH frames using TCP/22 which is to manage and initiate a session to the Memphis office e.g. from source IP 10.1.0.248 to destination IP 10.0.6.73. The transactions show encrypted payloads as the commands are not readable, it also shows small packets with ACKs which is a typical sign of admin sessions. As I mentioned previously there is a three-way handshake, so in this SSH client sends a SYN to server's port 22, server replies with SYN-ACK and the client returns ACK which establishes a connection to perform various admin tasks remotely, keeping the payload encrypted. Other opportunities include a Memphis employee needing to open a shared file or a

Dallas user needs to pull a billing report. These activities make the network busy with the traffic across WAN, so it is better that they are scheduled or prioritized to avoid real-time voice issues.

Patterns across the infrastructure - Apart from the SSH, MySQL and RTP, day to day traffic also includes Domain Name System (DNS, UDP/TCP 53), Dynamic Host Configuration Protocol (DHCP, UDP 67/68) for name resolution and addressing (typical/assumed day to day services, not shown explicitly in the capture). Session Initiation Protocol (SIP, 5060; TLS 5061) handles VoIP and call setup, it is recommended to use TLS on 5061 to secure and encrypt traffic. The most important protocol to access internet, cloud email/collaboration is Hypertext Transfer Protocol Secure (HTTPS, 443). Finally we have the Network Time Protocol (NTP, 123) which keeps the device clocks aligned for logs and troubleshooting. The essential network services like DHCP, DNS, NTP, and AD/LDAP run at all sites aiding in address assignment, name resolution, time synchronization and user authentication respectively.

Some of the observations I made while looking at the capture is that MySQL request/response pairs arrive in milliseconds making it low data-center latency. There is also steady use of RTP from provider to Memphis phone with packet sizes ~216 bytes, and shows sequence which indicates an active call. SSH session is very brief, which means a live or just initiated admin session, and light traffic with high privilege since it is an admin session. We can see internal communication (e.g. 10.0.8.73 to 10.0.8.42) for DB, and external to internal communication (67.16.104.172 → 10.0.6.73) for VoIP. MySQL is not using TLS which means that the DB traffic is unencrypted. These core network services (DNS/DHCP/SIP/NTP) are low bandwidth but critical and using them at each site and prioritizing RTP with QoS helps keep the voice table even if the usage grows.

Performance Issues - If we do not make any changes, the network will get to performance limits (bottleneck) very quickly because the Memphis office totally relies on Dallas's office internet and firewall. The growth of employees by 50% each year for the next two years will increase video/VoIP/cloud through the same edge and that will cause network congestion and choppy/jittery audio when data and calls overlap. The steady RTP flow in the capture shows how important this real-time-sensitive traffic is, so it is important that there are no issues with it. As mentioned in the milestone one, the Dallas office edge is a single-point-of-failure, any issue or even the maintenance window will impact Memphis office directly. No setup of QoS (priority for important tasks such as VoIP), updates, and backups will slow the network causing the call quality to degrade. As the employees are increased, wireless capacity will be impacted and network speed will take a hit. Currently the application to database transactions inside Dallas are rapid but the growth in users will increase query latency. Keeping milestone one feedback in mind, to add redundancy we can deploy dual routers per site (including new sites) and create primary and secondary LAN/WAN paths. Using managed switches with Virtual Local Area Networks (VLANs) and QoS (for RTP) and Power over Ethernet (PoE) for access points and phones will help mitigate these issues. VLAN is a logical subdivision in a network that will allow admins to segment a switch or network into multiple isolated networks. Each VLAN has its own broadcast domain, which basically means that devices in the same VLAN are able to communicate directly, and communication between different VLANs requires a router or a layer 3 switch. It is also used to improve security, performance, and to manage users by grouping the users/devices based on their functions, application, or department rather than actual physical location. In SNHUEnergy we will configure separate VLANs for HR, Accounting, IT and Finance departments on the same switch, but keeping the traffic secure and isolated. Separate

access of the internet on all sites will also make the offices independent which currently rely on the Dallas office. The implementation of automatic failover and load balancing between redundant WANs will aid in maintaining service during outages and evenly distribute traffic across various connections for better performance. This combined with automated response systems and real time network monitoring will help in preventing network congestion, detecting issues and rerouting traffic on need basis, which will ensure consistent data and voice quality as the company grows and usage scales.

Security Issues - If we do not improve security posture many issues will arise including firewall failure. The firewall/internet path at the Dallas office is a single point of failure that protects both sites. Failure or misconfiguration could expose or interrupt both offices. Minimal to no segmentation keeps all the devices on the same LAN, making it both security and performance risk. The Wireshark capture shows that MySQL is decoded (no TLS), which makes the DB traffic unencrypted that could be intercepted. SSH is highly privileged and should be restricted to management VLAN with multi-factor authentication. For voice, we should not use open SIP, but instead SIP-TLS (5061) should be used which makes the packets encrypted. The current model does not use any intrusion detection and intrusion prevention systems, and logging (SIEM), so the attacks and anomalies will go undetected. Keeping milestone one feedback in mind, the data should be encrypted at rest and in transit and a firewall should be deployed with IDS/IPS configuration to do strict monitoring of traffic and data. Some of the additional things we can implement to improve security are use of segmentation, enabling TLS on MySQL (3306), and SIP-TLS (5061) for voice. Creating a Demilitarized Zone (DMZ) for external applications will keep the public services such web or VPN isolated from the network internally, which reduces the exposure for external attacks. As discussed earlier, the implementation of RADIUS will add

an extra layer of access control and centralize the user verification. Keeping the network regularly backed up and having the restore procedure of important configurations always ready, ensures that the system is back up and running after a cyber incident or system failure.

Future Communication Needs - The company is expanding rapidly and for the company's system to run reliably, we need to make sure that the future communication needs are met to support this growth. I discussed the projected growth in the previous section, so keeping that in mind we must provide the offices with high bandwidth, low latency for essential functions such as video conferencing, VoIP, cloud collaboration, and data sharing. It is very critical that communication between the company headquarters' and the regional offices is reliable and secure to maintain workflow and simplified access to business resources for employees in all sites.

Focusing on the points above, the company should plan to implement hybrid cloud network infrastructure to complement the already existing on-prem infrastructure. According to Red Hat (n.d.), *“The term ‘cloud’ describes a network of on-demand computing resources accessed over the internet.”* Cloud based models give companies the flexibility to scale resources as the business grows without the heavy cost of new hardware. Red Hat (n.d.) further explains that *“Hybrid cloud refers to a combination of at least two computing environments that share information and run a uniform series of applications for a business or enterprise.”* I recommend using the combination of both on-prem and cloud services (hybrid model) which will help the company scale efficiently while reducing the cost and improving the access to applications such as payroll, email etc. from all the sites over the internet. For an inter-office file system I will recommend that the company should evaluate both the Network File System (NFS) hosted in Dallas and cloud based services such as Microsoft SharePoint. This reduces the need of having

costly on-prem file servers and it also provides secure and seamless access for employees as compared to on-prem NFS servers cloud repositories reduce the need of hardware, licensing costs and energy and will allow storage capacity to scale on demand. It will also strengthen the business continuity because ensuring that applications and critical data remains available in the cloud will make all the sites independent, because even if the headquarter site is down, the system will still be functional and same goes for other sites with applications. This model of hybrid cloud will help the company eliminate the single point of failure in Dallas. I will also recommend replicating applications and data in the cloud, this way all offices will have seamless access even during site outages. This helps us improve the business continuity and reduce downtime risks in the network. The day to day operations for SNHUEnergy depends on seamless data transfer between all sites, including future sites, which highlights the importance of WAN optimization, centralized resources access and redundancy. The future network infrastructure must implement centralized Active Directory (AD) and file storage, real time collaboration and remote connectivity using VPN and cloud services. The expansion to transportation and refining departments will increase the traffic in the network internally, to support that the company should implement segmentation and QoS prioritization to keep the network running securely and efficiently. To increase the uptime and business continuity we must also include automated failover systems, redundant connectivity, and network monitoring to monitor performance, anomalies, and intrusions so they are stopped before impacting the company operations. A well planned hybrid model combining on-prem and cloud services will help SNHUEnergy's growth in scalability, performance, and reliability to sustain business expansion. By looking at Wireshark packets, it confirms the company's critical flows such as MySQL (10.0.8.73 <> 10.0.8.42) for database queries and RTP (67.16.104.172 > 10.0.6.73) for VoIP, which indicates that the

SNHUEnergy's network supports hybrid communication between sites. This also validates that the company's network needs low latency WAN links, QoS prioritization, high bandwidth and encrypted database sessions for the future growth.

Network Architecture - As I discussed in the previous section, the hybrid network infrastructure model will perfectly align with SNHU Energy's business needs. This design will combine on-prem services (such as at headquarters in Dallas) with regional offices via Software-Defined Wide Area Network (SD-WAN) integration. As for cloud, I recalled a method which I learned while preparing for my cloud practitioner exam which was Hub-and-Spoke Topology where Dallas acts as a Hub (because it's the headquarter) and it connects to other offices (Spokes) such as Memphis, Houston, and Kansas City through the SD-WAN tunnels. According to an article published by Google Cloud on Medium (n.d.), *"The Hub-and-Spoke network topology is a model where all traffic flows along connections radiating out from a central node, or 'hub'. The hub is the central point of the network to which all other nodes, or 'spokes', are connected."* The model SD-WAN is to be operated over dual internet connections on all sites, with the use of secure cloud VPN gateways which will be connected to AWS or Azure. This will offer MPLS reliability without ongoing costs, which aligns with SNHUEnergy's budget for expansion. This method provides redundancy, cost efficiency, and smooth traffic routing while maintaining security and high performance.

Now I will discuss the actual physical devices we need to support this model. Firstly we will use the redundant routers and layer 3 switches using dynamic routing with Open Shortest Path First (OSPF) for automatic path selection. To distribute traffic we will implement firewalls, Remote Authentication Dial-In User Service (RADIUS) servers to control traffic and authenticate and Demilitarized Zones (DMZs) to separate public systems (VPN servers and web) from internal

systems. In order to strengthen the perimeter defense, redundant firewalls will be deployed at each office and the establishment of DMZ's to isolate public facing services like VPN and web gateways. As discussed in one the previous sections, the firewalls we will implement will have built in IDS and IPS to analyze, monitor and block malicious packets and activities in real time. Then we will add the devices, printers, and VoIP phones. We will implement VLANs to isolate and segment departments to increase security and performance, and also Dynamic Host Configuration Protocols (DHCP) which will assign IP addresses automatically. Another tool that will be implemented is enterprise applications such as ERP and CRM, and this will help in transitioning to Software as a Service (SaaS) model using platforms like Oracle NetSuite, Microsoft Dynamic 365, SAP HANA Cloud, or ZOHO CRM. This approach will eliminate the need of buying and maintaining the physical servers, it provides updates and vendor support continuously, and it also scales easily as the company grows. The existing on-prem servers in Dallas will keep on hosting core services such as DNS, DHCP, AD, and file management, meanwhile Memphis will still keep mirrored backups to ensure business continuity and redundancy. For storage performance and reliability, we can implement Storage Area Network (SAN) at the Dallas site. This will serve as a central, block level, and high speed storage system which will connect to the company's servers to host applications like ERP, SQL, and email systems. This will also allow fast access to shared storage, supports replication with cloud for disaster recovery and data backup is simplified. Hybrid approach by combining SAN with cloud services will ensure resilience against data loss during outages and high performance. To make the sites fault tolerant I will implement redundant firewalls and dual internet service provider (ISPs) network failover redundancy during outages. I will also implement load balancing to distribute the traffic evenly across the links to maintain performance and prevent

network congestion. Focusing on logical network services I will implement DHCP, DNS, Network Time Protocol (NTP) for time sync and Lightweight Directory Access Protocol (LDAP) to manage users which will be hosted in Dallas and will be mirrored in Memphis for redundancy. For security I will implement Security Information and Event Management (SIEM), there are various tools such as Splunk, Microsoft Sentinel which are used for this and to monitor traffic in the network. Regularly backing up data and replicating the cloud will also be implemented to ensure quick recovery after an incident. All of these things will make SNHUEnergy's network secure, scalable and resilient while also supporting the future expansion of the organization. All sites future design will include redundancy at both the LAN and WAN layers. This will ensure that routers, switches, and Internet can fail over automatically during outages without disrupting communication. After the migration, the legacy servers and storage will be decommissioned following company procedures or repurposed for testing. This will also reduce the power usage and maintenance cost meanwhile the resources are redirected to the cloud. All the offices will have dual routers and layer 3 switches configuring primary and secondary LAN for load balancing and automatic failover. This creates redundancy links and eliminates the risk of single point of failure and maintaining performance and availability during incidents and outages. The Wireshark packets that are captured like MySQL, RTP and SSH, demonstrates inter-site dependency. But the proposed model with SD-WAN and dual ISPs will ensure QoS prioritization and consistent latency for critical flows.

IP Addressing and Subnets - This section outlines the IP addressing and subnetting strategy designed to ensure manageability, scalability, and organization within the enterprise network. SNHUEnergy will use a private 10.0.0.0/16 range, which provides 65,534 usable host addresses, which will be divided into /24 subnets for each department and site. This structure enhances

isolation between departments, simplifies network management, and provides sufficient capacity for future growth.

Dallas (Hub/HQ) - Subnet 10.0.10.0/24 - Assigned to core infrastructure and application servers supporting critical services such as AD, DNS, DHCP, and File Services.

Departmental VLANs - VLAN 10 - HR & Finance: 10.0.11.0/24 & VLAN 20 - IT & Network Management: 10.0.12.0/24. These provide logical segmentation between administrative and technical departments, ensuring improved performance, security, and access control.

Branch Offices (Spokes) - Memphis - subnet 10.0.20.0/24 - Operations and Billing. Kansas City - subnet 10.0.30.0/24 - Transportation and Logistics. Houston - subnet 10.0.40.0/24 - Refining and Field Operations. Each site operated independently while remaining securely interconnected via the WAN.

VPN and WAN Links - SD-WAN Tunnels: 10.0.100.0/30 & Remote Access VPN Users: 10.0.200.0/24. These provide secure connectivity for remote users and also facilitates site to site communication between branch locations and data centers.

DHCP will handle dynamic IP assignments, while static IPs will be reserved for critical devices such as servers, routers, and switches. To ensure high availability and business continuity, a redundant (failover) DNS and DHCP configuration will be implemented between the Dallas and Memphis offices. This addressing plan will ensure scalability by providing enough capacity for future growth of each department and the new offices. Logical segmentation will limit broadcast domains and improve security through role-based separation, which will ensure that sensitive data is protected within department boundaries. The /24 subnetting simplifies routing, DHCP management, and daily troubleshooting which improves overall manageability. It also complies with RFC 1918 standards and follows enterprise best practices to maintain a secure network. The

subnet has also been intentionally spaced to allow future expansion, establishing new VLANs or adding new departments without overlap, or major reconfiguration.

Planning and Security

Mitigating Performance and Security Issues - The proposed network design is emphasized on segmentation, redundancy, and proactive defense. The safeguards discussed are not arbitrary, they directly address weaknesses which I identified in earlier analysis including lack of visibility, single point of failures and unsecured communication internally. Implementing redundant WAN links and dual routers will ensure business continuity if primary LAN or device fails. This layering of redundancy shows a defense in depth model where availability and security coexist. Performance will be optimized via VLAN segmentation and QoS prioritization will further protect critical operations such as voice and video as the demand grows. Instead of adding bandwidth, these policies intelligently prioritize traffic and isolate domains, reducing congestion and latency. The combined implementation of continuous monitoring, load balancing and automatic failover forms a model which can self heal and prevent minor issues from making them enterprise wide disruptions. Security is improved via reactive and preventive controls, where firewalls are paired with IDS/IPS to stop unauthorized access and situational awareness by behavioral inspection. These plans will help the company's network and instead of reactive actions, it will create predictive stability and controlled exposure. I have also discussed solutions in my Security Issues section. When I mentioned them I also provided ways to mitigate the issues.

Network Management Tool - In an enterprise network visibility and automation is very crucial and is thought of as backbone for sustainability. For that it is important we implement centralized management platforms like PRTG Network Monitor or SolarWinds Network Performance

Monitor and integrating it with SIEM (Splunk or Microsoft Sentinel) will provide necessary insights for reliability. These are not just for metrics but also to correlate events, issue actionable alerts real time and detect anomalies. The administrators can trace performance degradation to specific issues, identify early signs of compromise and anticipate bandwidth bottleneck by combining telemetry with security event data in a network. This proactive approach helps with data driven decision making, justify budgets, allows IT admins to plan upgrades, and ensures compliance without manual guesswork. These tools have a goal that a network is measured, monitored and improved continuously. Additionally, the Simple Network Management Protocol (SNMP) will also be implemented across all network devices like switches, routers, firewalls and servers. The dedicated SNMP agents continuously monitor and collect real time data such as interface errors, bandwidth usage, device uptime and this information is then sent to central monitoring platforms. This provides network administrators early visibility into link or hardware issues. Enabling SNMP with tools like SolarWinds or PRTG provides a single dashboard for trend analysis, proactive monitoring and automated alerts for fast troubleshooting and consistent performance.

Security Devices and Integration - The proposed security controls like NGFW with IDS/IPS configurations, DMZ segmentation and RADIUS authentication ensures secure architecture built on least privilege model. Each mitigates its unique risk vector, RADIUS centralizes access control, DMZs isolates public facing services from internal network, firewalls protect boundaries, and IDS/IPS ensures deep packet inspection. This model complements existing controls instead of replacing them. Combining RADIUS with AD enforces SSO on both on-prem and cloud services leaving no gaps in credential management. Deploying the security applications at each site guarantees uniform policy and reduces dependency on single site.

Additionally, this also extends the company's zero-trust posture and prepares a network for compliance with frameworks such as NIST CSF and ISO 27001.

Changes to Existing Devices - Implementing these new technologies does not replace but requires refinement. The current switches and routers need to support Open Shortest Path First (OSPF) routing, QoS prioritization and VLAN tagging, aligning them with enterprise hardware. Firmware needs to be updated and applying standardized configurations are key to ensure compatibility with VPN and SD-WAN tunnels. The current servers need to incorporate LDAP and RADIUS for centralized authentication. DNS and DHCP will be used for seamless address resolution. These improvements preserve previous expenditure, aligning legacy systems with modern tools and services for security and automation to enforce the company's policy of continuous betterment without major disruptions.

Implementation Challenges - The transition from static to dynamic network environment has its challenges some of which are predictable, like configuration drift, service interruptions, and learning curve of all the staff specially the technical staff and admins. Key strategy is to phase out the transition, meaning new technology and services should be done using phased deployment, robust change control and non-peak/after hours cutovers. Each new feature (such as redundancy, IDS/IPS) should be pilot tested before at a single site which will reveal any issues related to interoperability. User adaptation is very critical, this can be done using training sessions for IT admins on various tools such as incident triage, SIEM, and network analytics which will help us understand if the new tools are delivering the value intended. The communication between business units during transitions keeps transparency and maintains trust, to reduce perceived downtime or complexity.

Overall Risk of Not Maintaining Security Standards - Upkeep of security is very important, neglecting it will erode every feature and design advantage that is built into the network. For example, expired IDS signature, patching lapse or unmonitored SIEM alert could make a minor issue or vulnerability into full scale breach. When it comes to operations, these mean policy and compliance violations, data exposure and loss of confidence from customers. Setting up regular update cycles, penetration testing, and scheduled audits, ensures and checks continuous alignment alongside evolving threats. Actual risk is not in initial deployment because we pilot tested before deploying but in complacency afterwards. Security needs continuous discipline, maintaining it is crucial and fundamental to integrity, performance and reputation of any organization.

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